

Wholesale Customer Segmentation

Executive Summary & Decision Framework

Executive Summary

Problem: Identify meaningful wholesale customer segments from annual spending across six product categories to support evidence-based commercial hypotheses.

Method: Log1p transformation, StandardScaler, K-Means clustering with K evaluated from 2 to 10, hierarchical clustering as a cross-check, multiple-seed consistency, bootstrap stability, and business interpretability.

Final K: 2. K=2 had the highest K-Means silhouette among stability-qualified candidates (0.290) and also the strongest hierarchical silhouette (0.258). Bootstrap mean ARI was 0.92 and seed-variation mean ARI was 1.00.

Cluster sizes: Cluster 0 = 252 customers (57.3%); Cluster 1 = 188 customers (42.7%).

Key difference: Cluster 0 is dominated by Fresh and Frozen spending and is 96.8% Horeca. Cluster 1 is dominated by Grocery and Milk spending and is 71.3% Retail. Mean total annual spend is 24,529 versus 44,884 respectively.

Business meaning: The segmentation is best used as a decision-support framework. Recommendations below are testable hypotheses, not causal claims.

Model Selection

K	Silhouette	Bootstrap ARI	Stability	Decision
2	0.290	0.92	Strong	Selected
3	0.258	0.79	Moderate	Rejected
4	0.188	0.62	Weak	Rejected

Decision rule: select the highest K-Means silhouette among candidates that pass both seed-variation and bootstrap stability checks. The project-defined ARI threshold is 0.75.

Decision Framework

Cluster 0 — Horeca Fresh/Frozen Buyers 252 customers (57.3%).	Prioritize Fresh/Frozen product bundles. Focus on Horeca-specific offers. Monitor high-volume accounts and order frequency when those measures become available. Evidence: Fresh mean spend 13,973; Frozen 3,706; 96.8% Horeca.
Cluster 1 — Retail Grocery/Milk Buyers 188 customers (42.7%).	Prioritize Grocery/Milk cross-selling. Develop retail-oriented promotions. Identify high-value customers within the segment using spend or margin once available. Evidence: Grocery mean spend 14,697; Milk 10,346; 71.3% Retail.

Guardrail: These recommendations describe business hypotheses supported by observed category and channel patterns. They do not establish that a promotion, pricing action, delivery change, or other intervention will cause a sales outcome.

1. Introduction & Dataset Overview

This project applies unsupervised clustering to segment wholesale customers by annual spending across six product categories. The objective is to identify purchasing-pattern segments that can support business hypotheses around promotions, service design, inventory planning, and customer management. K-Means is the primary method, while Ward hierarchical clustering is used as a methodological cross-check.

Dataset source: [UCI Machine Learning Repository — Wholesale Customers](#)

Variable	Meaning	Role
Channel	Distribution channel: 1 = Horeca, 2 = Retail	Post-hoc profiling
Region	Region code: 1 = Lisbon, 2 = Oporto, 3 = Other	Post-hoc profiling
Fresh	Annual spending on fresh products	Clustering
Milk	Annual spending on milk products	Clustering
Grocery	Annual spending on grocery products	Clustering
Frozen	Annual spending on frozen products	Clustering
Detergents_Paper	Annual spending on detergents and paper	Clustering
Delicassen	Annual spending on delicatessen products	Clustering

2. Data Overview & Quality

The dataset contains 440 customer records across 8 columns. Core quality checks found 0 missing values, 0 duplicate rows, 0 negative spending values, and 0 zero spending values across the six clustering variables. No constant columns were found.

Channel distribution (1 = Horeca, 2 = Retail):

Channel	Count
1	298
2	142

Region distribution (1 = Lisbon, 2 = Oporto, 3 = Other):

Region	Count
1	77
2	47
3	316

Univariate EDA: Spend Category Distributions

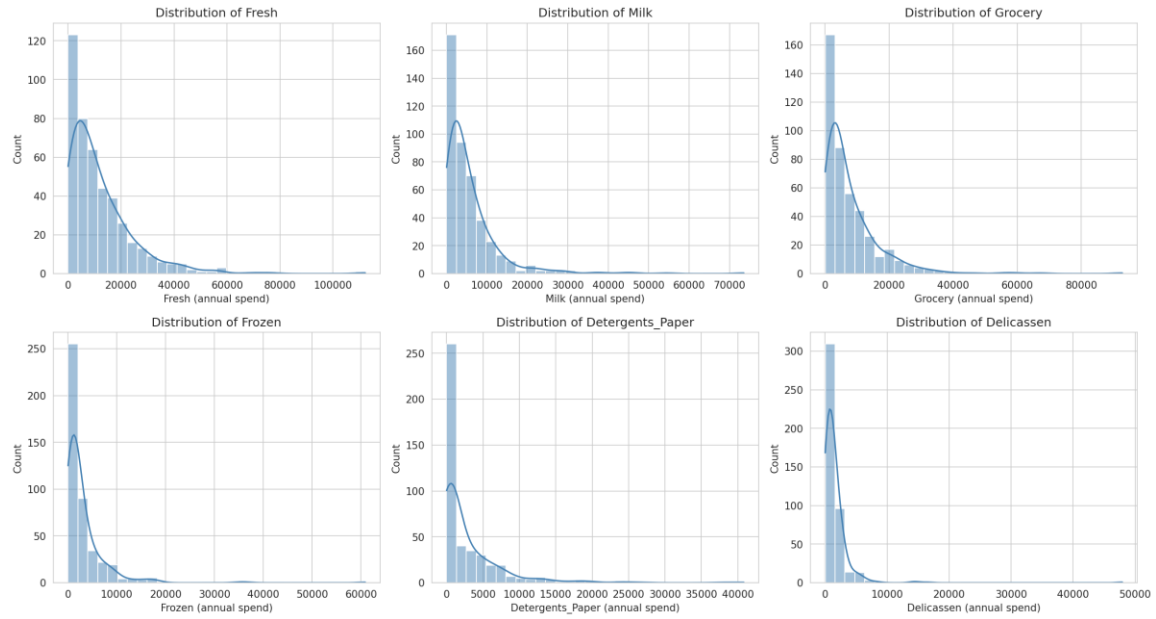


Figure 1. Annual spending distributions across the six clustering categories; all are strongly right-skewed.

Univariate EDA: Spend Category Boxplots (Outlier View)

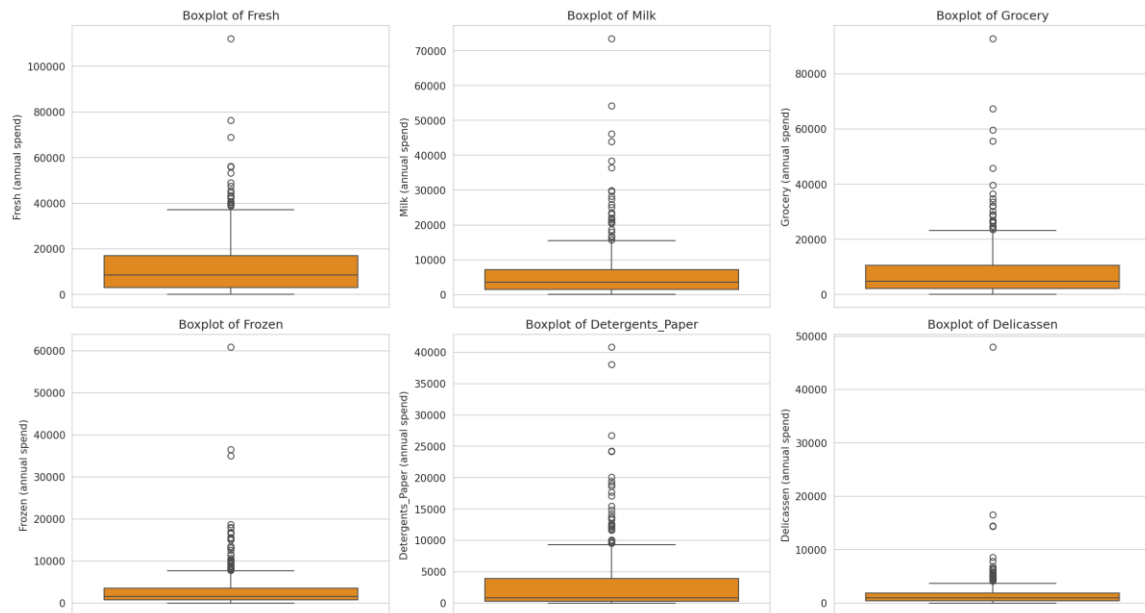


Figure 2. Boxplots showing spread and potential high-spend observations.

Univariate EDA: Categorical Field Distributions

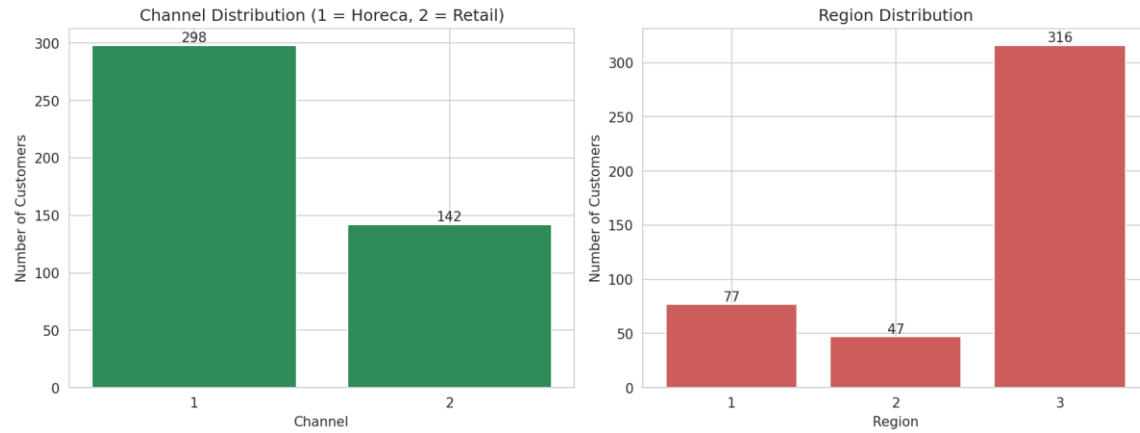


Figure 3. Distribution of Channel and Region categories.

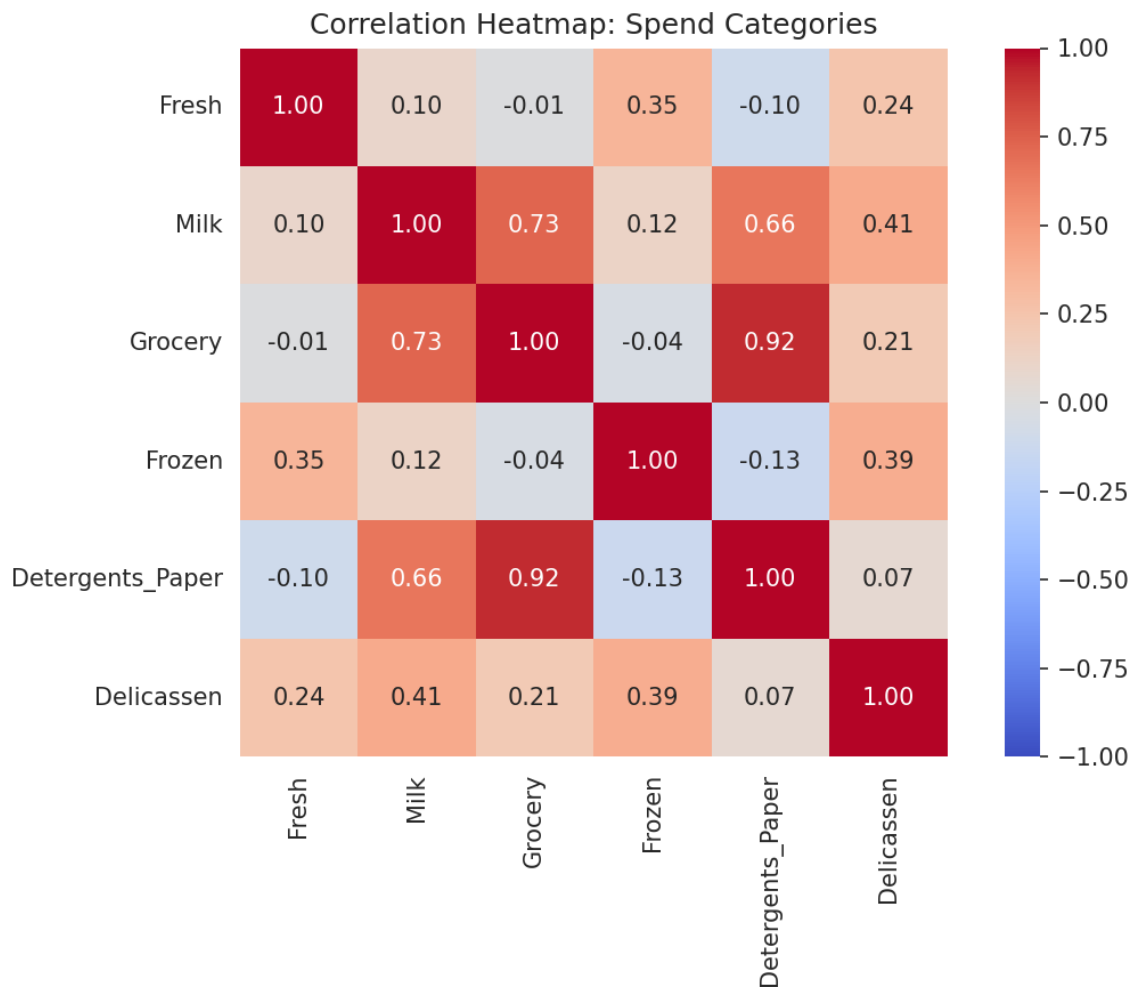


Figure 4. Correlation structure among the six spending variables.



Figure 5. Exploratory pairplot colored by Channel; Channel is not used as a clustering input.

3. Methodology

3.1 Feature Selection & Redundancy Diagnostics

The six spending variables were used as the clustering feature set because they directly describe purchasing behavior. Channel and Region were excluded from clustering and reserved for post-hoc profiling. Correlation and VIF were examined as redundancy diagnostics. Because the spending categories are individually business-relevant, no feature was dropped. PCA was reserved for visualization rather than used as a substitute for the six-feature K-Means input.

Feature	VIF
Fresh	1.18
Milk	2.55
Grocery	8.79
Frozen	1.32
Detergents_Paper	8.1
Delicassen	1.52

3.2 Skewness, Outliers & Treatment

All six spend variables are strongly right-skewed. IQR screening identified potential extreme observations, but these were retained because the available evidence does not establish them as data errors and high-spend customers are substantively relevant to segmentation. A log1p transformation was applied to compress extreme values and reduce skew. The transformation was evaluated by comparing skewness before and after it.

Feature	Skew Before	Skew After	IQR Outliers
Fresh	2.56	-1.58	20
Milk	4.05	-0.22	28
Grocery	3.59	-0.67	24
Frozen	5.91	-0.35	43
Detergents_Paper	3.63	-0.24	30
Delicassen	11.15	-1.09	27

Outlier Reference: Boxplots Annotated with IQR Outlier Counts

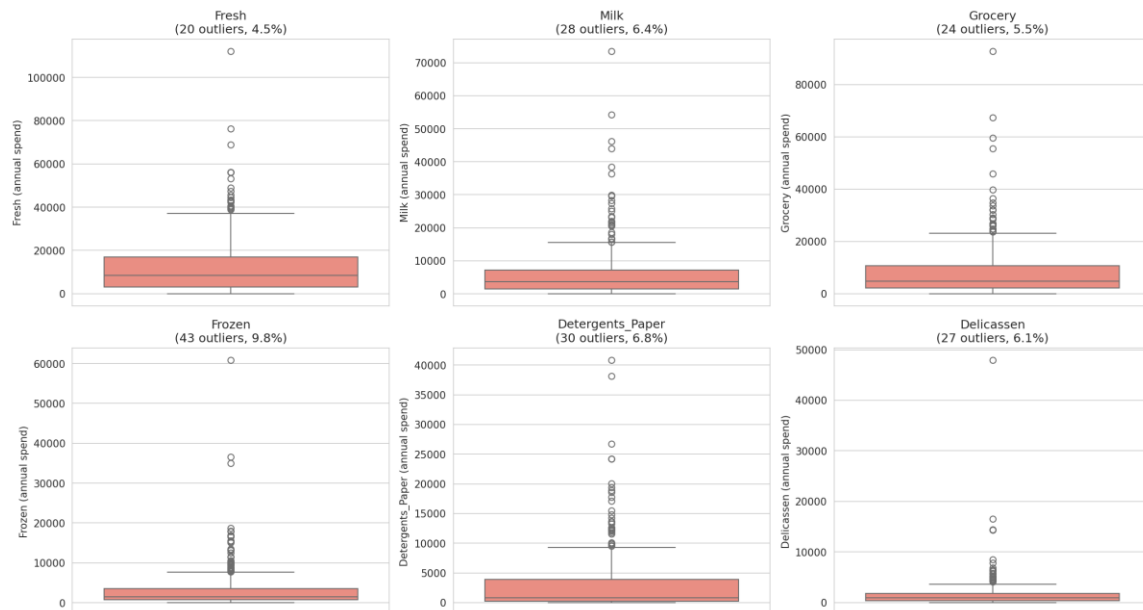


Figure 6. IQR-based outlier counts by spending category.

Effect of log1p Transform on Spend Distributions

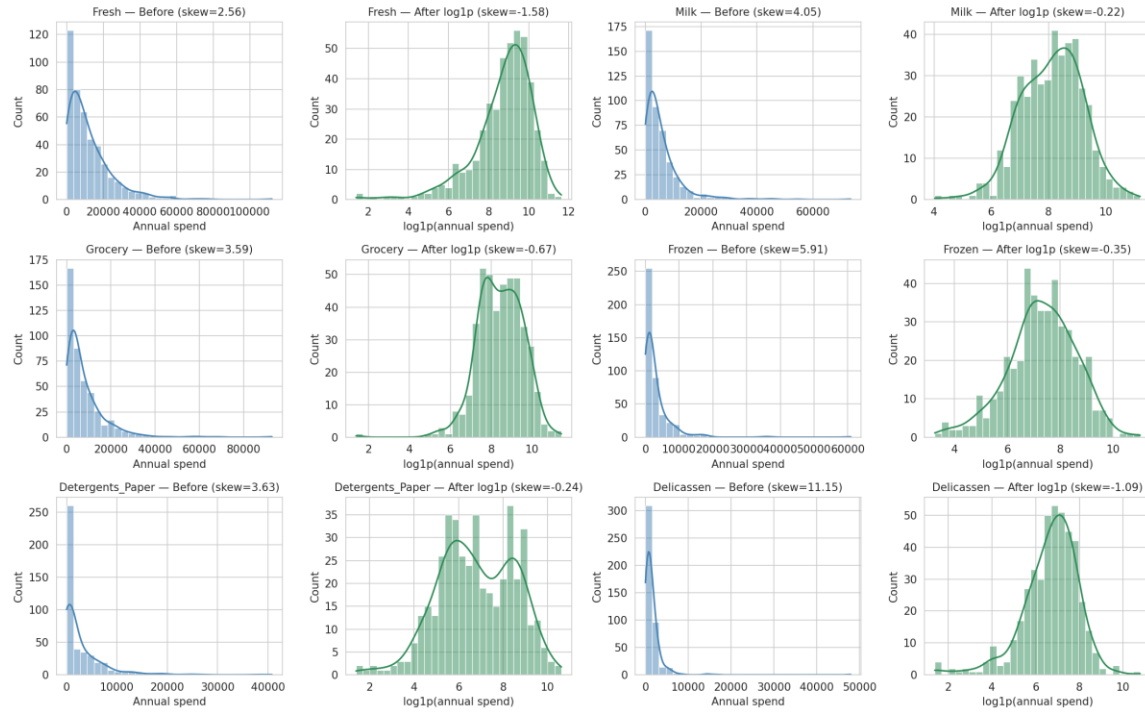


Figure 7. Effect of log1p on the six spending distributions.

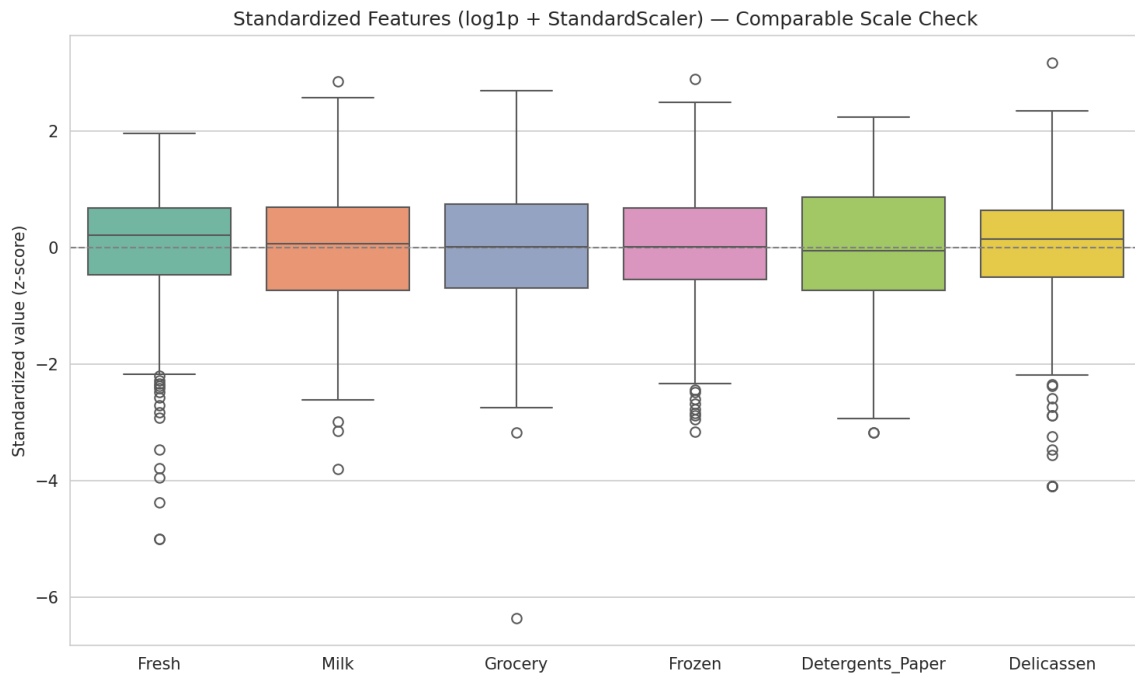


Figure 8. Final standardized feature distributions.

Code: log1p transformation applied to the six spending variables.

```
log_df = features_df.copy()
for col in SPEND_COLS:
    log_df[col] = np.log1p(features_df[col])
```

3.3 Scaling

After transformation, StandardScaler was applied so each clustering feature contributed on a comparable standardized scale. This is important because K-Means minimizes squared Euclidean distances and is therefore sensitive to feature scale.

Code: StandardScaler fitted to the log-transformed feature matrix.

```
scaler = StandardScaler()
scaled_array = scaler.fit_transform(log_df[SPEND_COLS])
scaled_df = pd.DataFrame(scaled_array, columns=SPEND_COLS, index=log_df.index)
```

3.4 Algorithm Selection & Number of Clusters

K-Means and Ward hierarchical clustering were evaluated for K=2 through K=10. K-Means achieved its highest average silhouette at K=2 (0.2903). The inertia curve does not show a uniquely sharp elbow, so the elbow plot is treated as supporting evidence only rather than as an automated selection rule. The hierarchical model's highest silhouette also occurred at K=2. The final choice therefore gives greater weight to silhouette separation, clustering stability, cross-method comparison, and business interpretability.

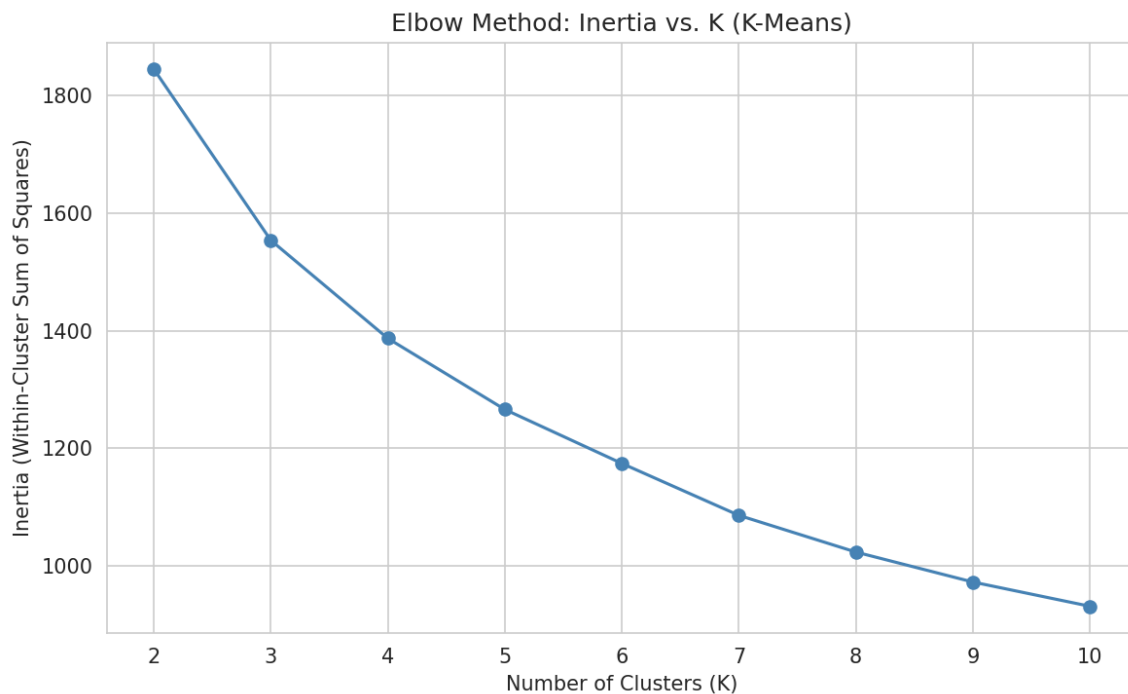


Figure 9. K-Means elbow curve across K=2-10.

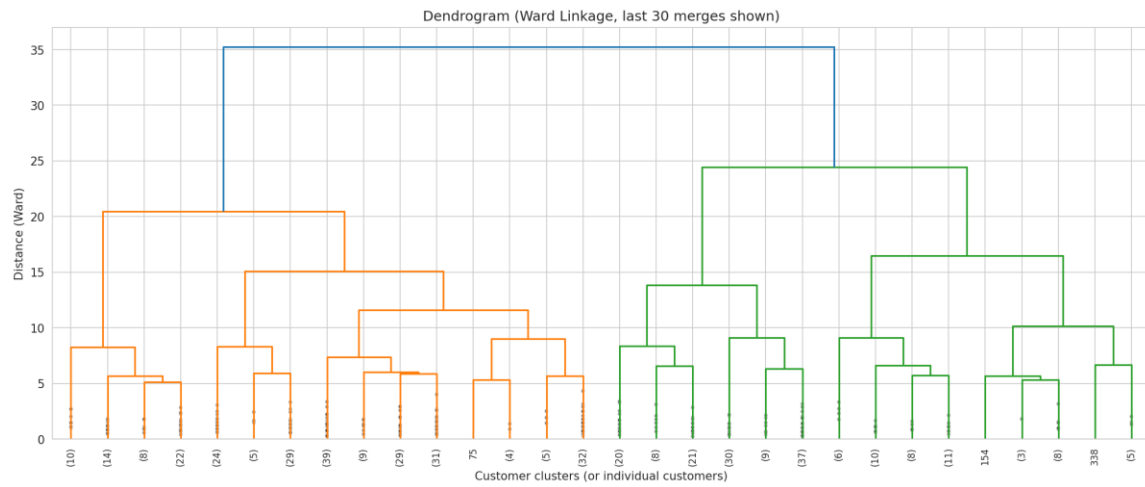


Figure 10. Ward hierarchical dendrogram used as a qualitative structural cross-check.

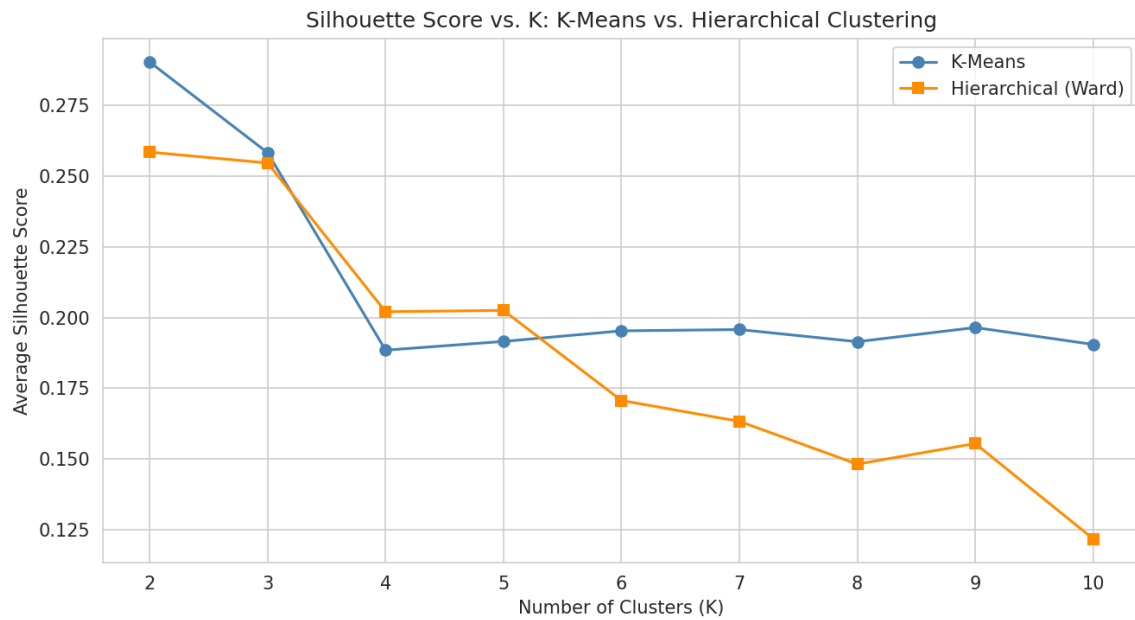


Figure 11. Average silhouette scores for K-Means and hierarchical clustering.

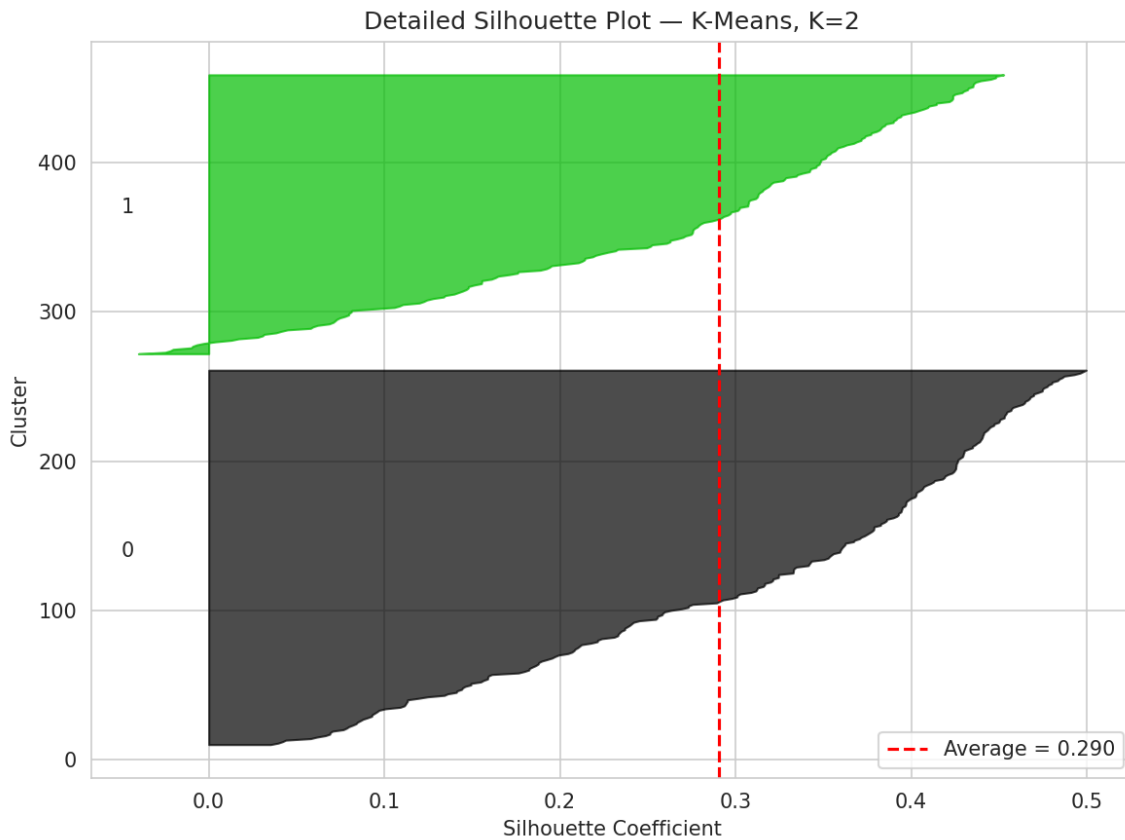


Figure 12. Detailed silhouette profile for the selected K=2 solution.

K	K-Means Inertia	K-Means Silhouette	Hierarchical Silhouette
2	1844.06	0.2903	0.2585
3	1553.41	0.2583	0.2547
4	1386.86	0.1885	0.2021
5	1266.15	0.1916	0.2025
6	1174.54	0.1953	0.1707
7	1086.36	0.1958	0.1633
8	1024.05	0.1915	0.1482
9	973.14	0.1965	0.1555
10	932.00	0.1905	0.1218

Code: K-Means evaluation across K=2-10 using inertia and silhouette score.

```
for k in K_RANGE:
    km = KMeans(n_clusters=k, random_state=RANDOM_STATE, n_init=N_INIT)
    labels = km.fit_predict(X)
    sil = silhouette_score(X, labels)
    rows.append({"k": k, "inertia": km.inertia_, "silhouette": sil})
    print(f" K={k:2d} inertia={km.inertia_:10.2f} silhouette={sil:.4f}")
```

3.5 Final Model & Stability Validation

The primary K-Means model uses K=2, random_state=42, and n_init=20. Five random seeds and 50 bootstrap resamples were used to test whether the clustering was sensitive to initialization or sample variation. K=2 had seed-variation ARI=1.00 and bootstrap ARI=0.92; both exceed the project-defined heuristic stability threshold of 0.75. K=3

and K=4 were also checked because they are plausible more-granular alternatives. K=3 had seed ARI=0.98 and bootstrap ARI=0.79; K=4 had seed ARI=1.00 and bootstrap ARI=0.62.

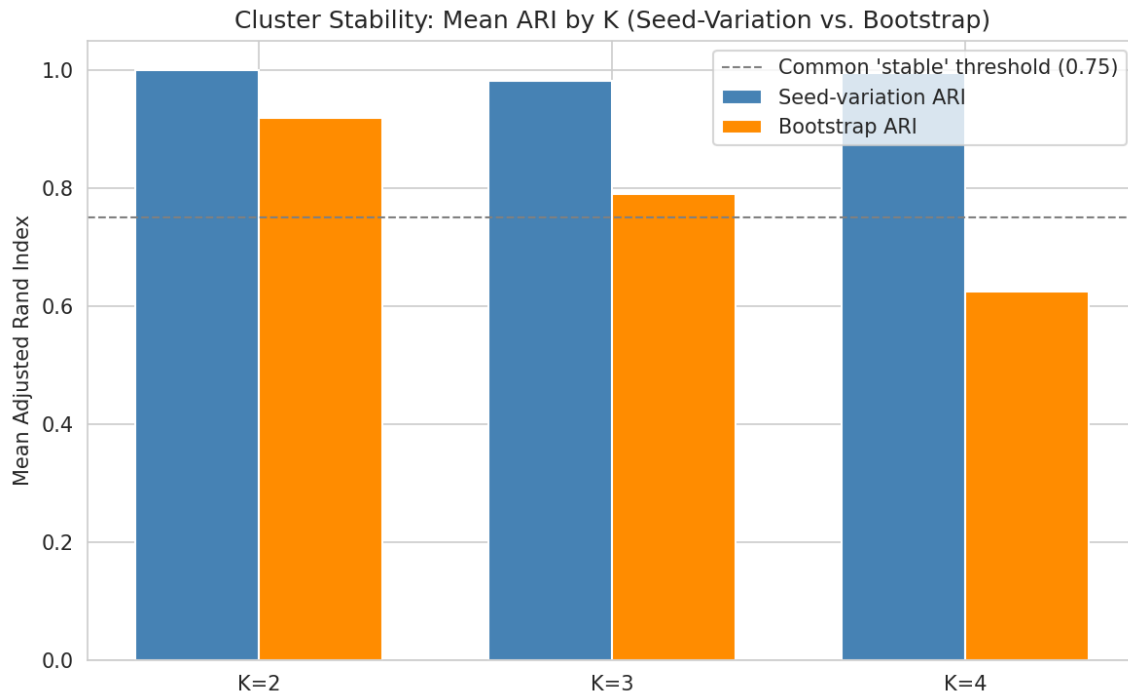


Figure 13. Mean stability scores for K=2, K=3, and K=4.

K	Seed ARI	Bootstrap ARI	Verdict
2	1.00	0.92	Stable
3	0.98	0.79	Stable
4	1.00	0.62	Not stable

Code: final K-Means fitting with fixed reproducibility settings.

```
for k in CANDIDATE_KS:
    km = KMeans(n_clusters=k, random_state=RANDOM_STATE, n_init=N_INIT)
    labels = km.fit_predict(X)
    models[k] = {"model": km, "labels": labels}
    sizes = pd.Series(labels).value_counts().sort_index()
    print(f"\nK={k}: trained. Cluster sizes: {sizes.to_dict()} "
          f"(inertia={km.inertia_:.2f})")
```

Code: Adjusted Rand Index stability check across random seeds.

```
for seed in STABILITY_SEEDS:
    km = KMeans(n_clusters=k, random_state=seed, n_init=N_INIT)
    seed_labels[seed] = km.fit_predict(X)

rows = []
seeds = list(seed_labels.keys())
for i in range(len(seeds)):
    for j in range(i + 1, len(seeds)):
        ari = adjusted_rand_score(seed_labels[seeds[i]], seed_labels[seeds[j]])
        rows.append({"seed_a": seeds[i], "seed_b": seeds[j], "ari": ari})
```

4. Results

4.1 Final Cluster Profiles

Cluster	Count	% of Total
Cluster 0	252	57.3%
Cluster 1	188	42.7%

Cluster	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	13,973.1	2,401.8	2,918.7	3,705.7	491.9	1,038.0
1	9,355.9	10,346.4	14,697.1	2,222.5	6,084.5	2,177.4

Cluster	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	10,347.5	1,871.0	2,406.0	2,225.0	327.5	749.5
1	5,406.5	7,690.5	11,527.0	1,061.5	4,495.0	1,388.5

Feature	F-statistic	Eta-squared
Grocery	264.79	0.377
Detergents_Paper	223.09	0.337
Milk	173.91	0.284
Delicassen	18.27	0.040
Fresh	14.80	0.033
Frozen	10.26	0.023

Interpretation note: F-statistic and eta-squared are used descriptively because the same spending variables were used to construct the clusters; their p-values are not treated as independent confirmation.

Cluster	Horeca	Retail
0	96.8%	3.2%
1	28.7%	71.3%

Cluster	1	2	3
0	19.0%	11.1%	69.8%
1	15.4%	10.1%	74.5%

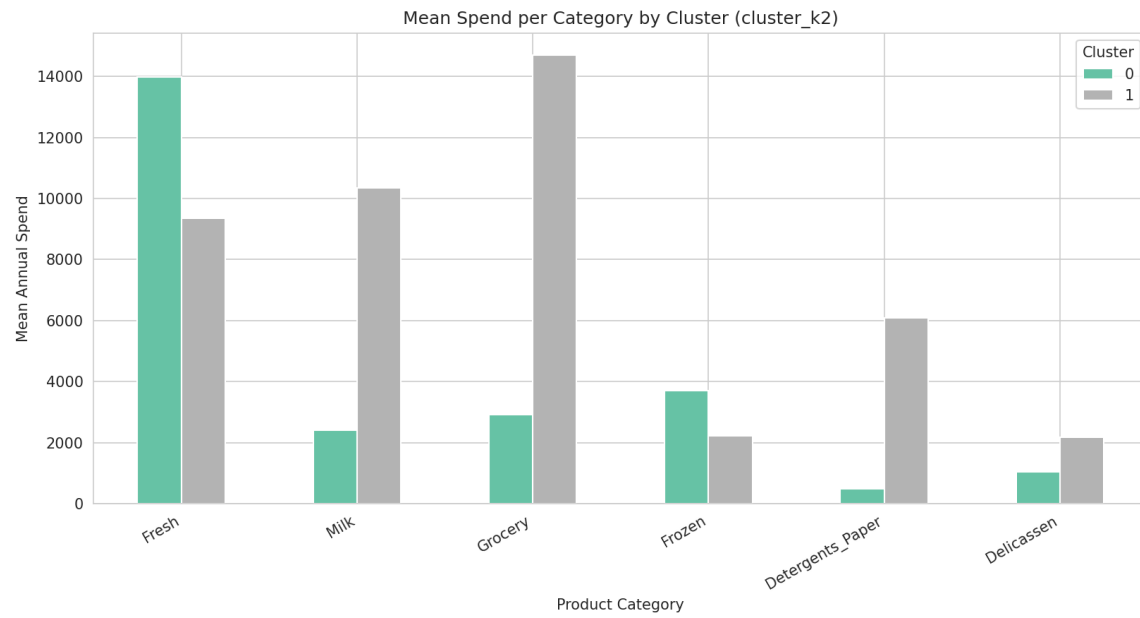


Figure 14. Mean annual spend by product category and cluster.

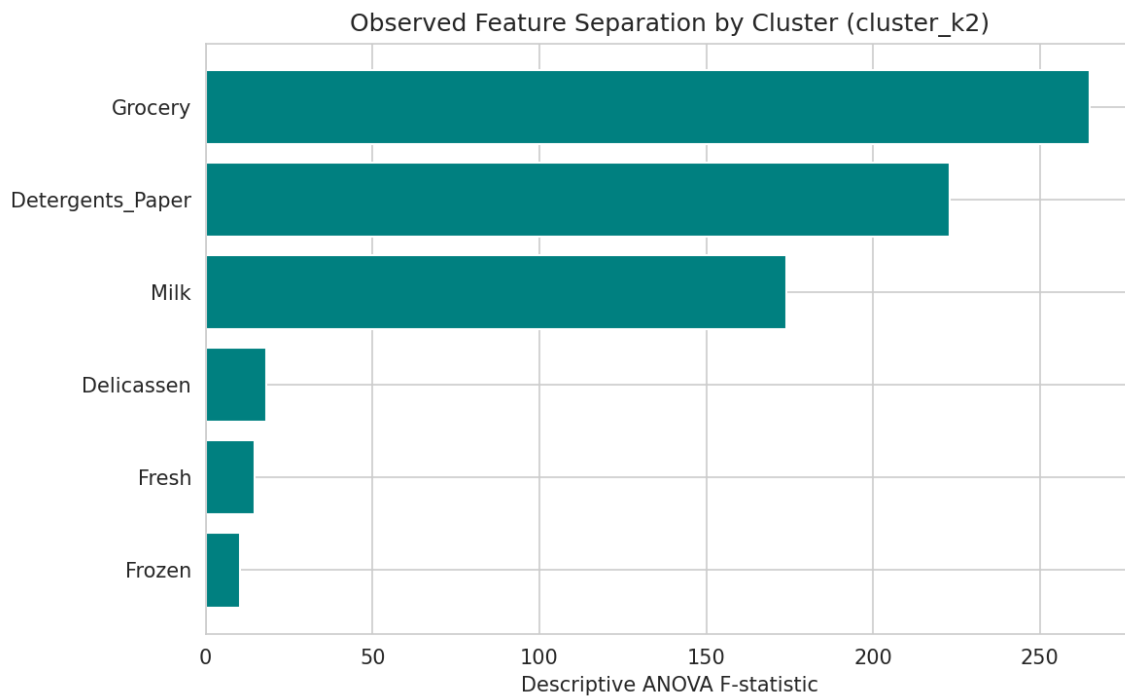


Figure 15. Descriptive ANOVA F-statistics ranking observed feature separation.

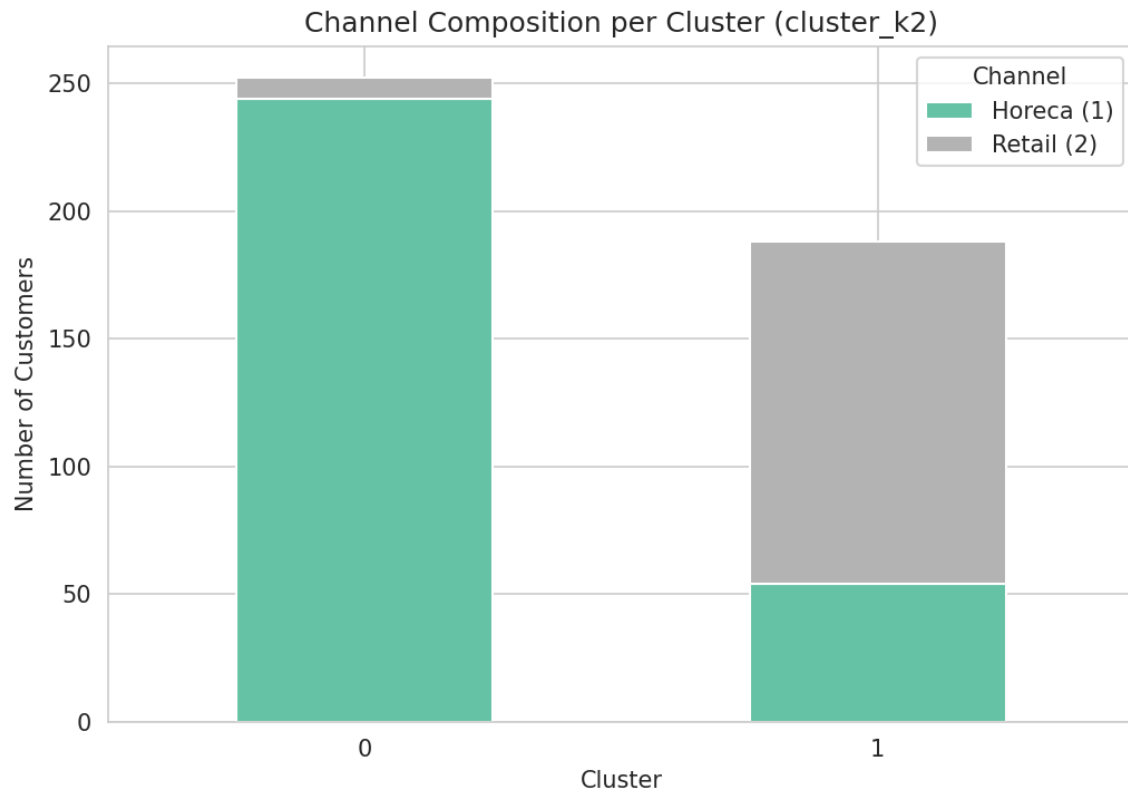


Figure 16. Channel composition within the discovered clusters.



Figure 17. Region composition within the discovered clusters.

Code: cluster labeling and retention of original, human-readable spend values for profiling.

```
for k in CANDIDATE_KS:
```

```

km = KMeans(n_clusters=k, random_state=RANDOM_STATE, n_init=N_INIT)
labels = km.fit_predict(X)
models[k] = {"model": km, "labels": labels}
sizes = pd.Series(labels).value_counts().sort_index()
print(f"\nk={k}: trained. Cluster sizes: {sizes.to_dict()} "
      f"(inertia={km.inertia_:.2f})")

```

4.2 PCA Visualization

PCA reduces the six-dimensional standardized feature space to two components for visualization only. PC1 explains 44.1% of the variance and PC2 explains 27.2%, for a combined 71.3%. The PCA plot is a two-dimensional projection and is not the basis for the clustering itself.

Feature	PC1	PC2
Fresh	-0.105	0.590
Milk	0.542	0.133
Grocery	0.571	-0.006
Frozen	-0.138	0.590
Detergents_Paper	0.551	-0.071
Delicassen	0.214	0.530

PCA Component Loadings by Feature

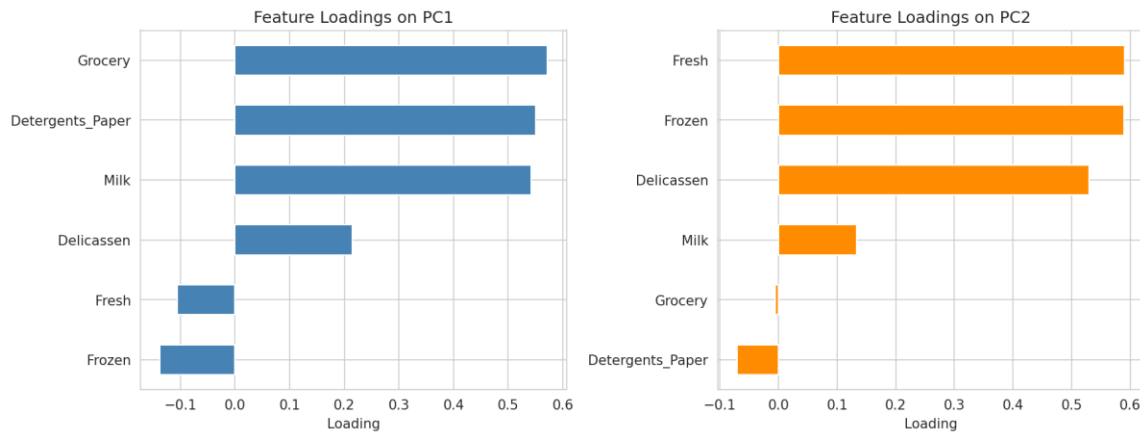


Figure 18. Feature loadings on PC1 and PC2.

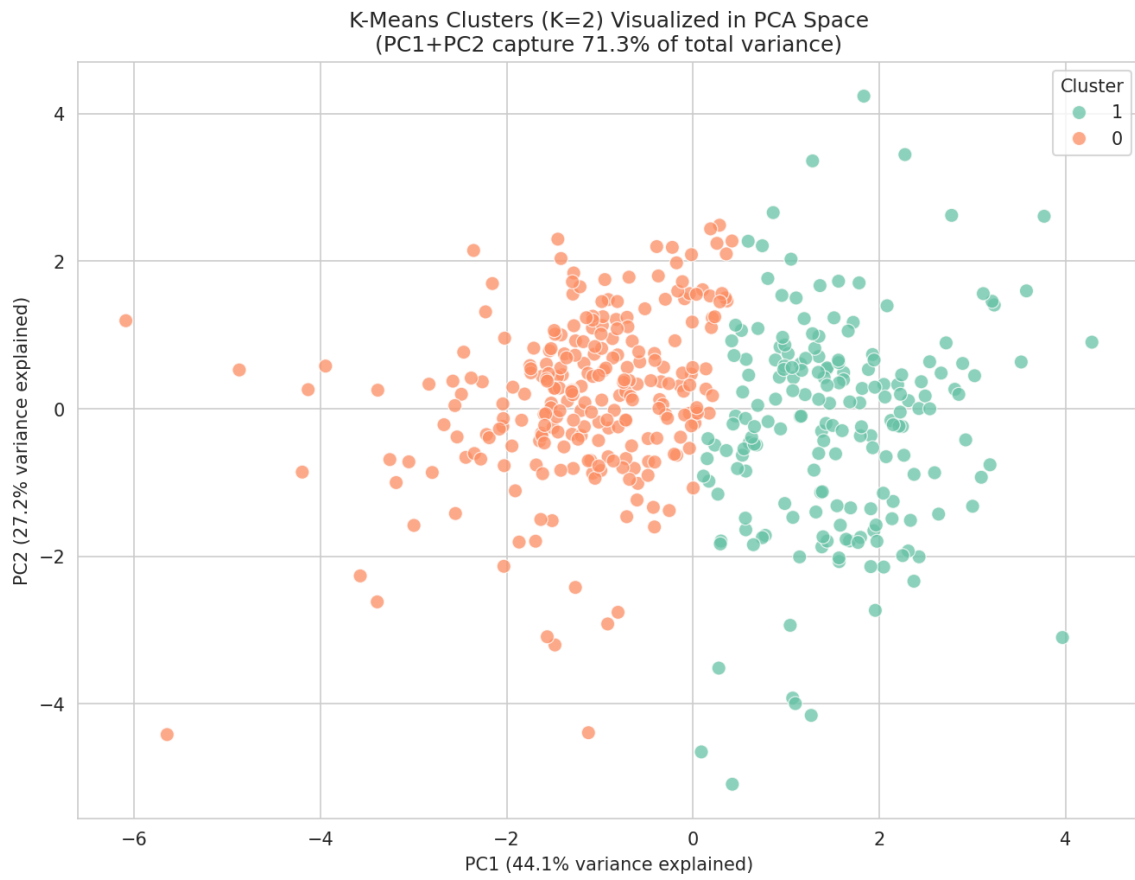


Figure 19. K=2 clusters projected into PCA space.

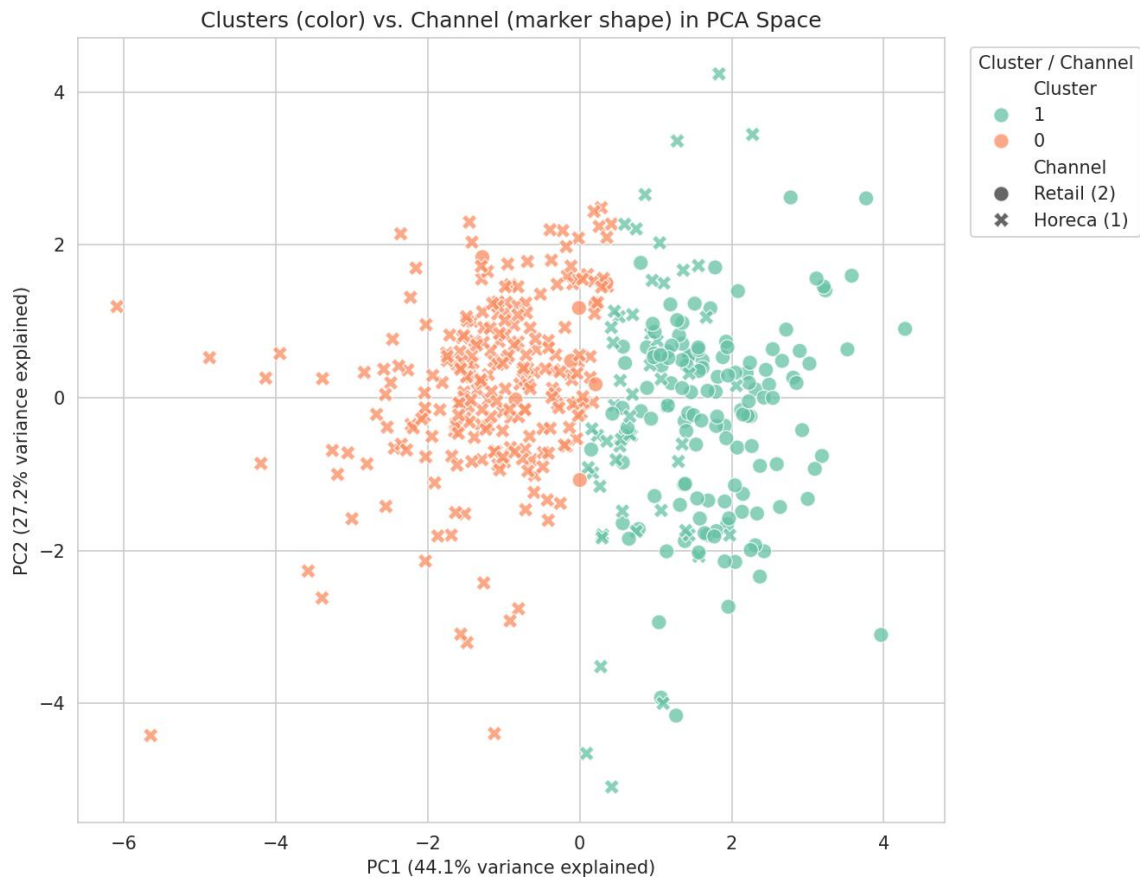


Figure 20. Cluster membership and Channel shown together in PCA space.

Code: PCA used solely to create a two-dimensional diagnostic visualization.

```
pca = PCA(n_components=2, random_state=RANDOM_STATE)
pcs = pca.fit_transform(scaled_df[SPEND_COLS])
```

5. Cluster Interpretation

Cluster 0 — Lower-Total-Spend Fresh/Frozen-Oriented Segment

Size: 252 customers (57.3%).

Highest mean-spend categories: Fresh (13,973) and Frozen (3,706).

Mean total annual spend: 24,529.

Post-hoc Channel profile: 96.8% Horeca.

Cluster 1 — Higher-Total-Spend Grocery/Milk-Oriented Segment

Size: 188 customers (42.7%).

Highest mean-spend categories: Grocery (14,697) and Milk (10,346).

Mean total annual spend: 44,884.

Post-hoc Channel profile: 71.3% Retail.

Cluster 1 has the higher mean total annual spend (44,884) and is strongest in Grocery and Milk. Cluster 0 has the lower mean total annual spend (24,529) and is strongest in Fresh and Frozen.

6. Business Implications

Targeted promotions: Cluster 1's Grocery emphasis supports testing category-bundled offers. Cluster 0's Fresh/Frozen-oriented profile supports testing more category-specific offers.

Service and inventory planning: Cluster 0 is strongly Horeca-dominant and Fresh/Frozen-heavy. This supports a hypothesis around perishable-focused service options or different delivery frequencies, but delivery frequency is not contained in this dataset and must be validated separately.

Commercial strategy: Cluster 1 has higher mean annual spend (44,884 vs. 24,529), supporting a hypothesis for testing volume-based pricing or loyalty structures. These are decision-support hypotheses, not causal findings.

7. Limitations

- 1. Sample size:** 440 customers is adequate for an educational clustering analysis but still limits external generalization.
- 2. Time aggregation:** the variables are annual spending totals, so seasonality and customer movement between segments cannot be observed.
- 3. Feature scope:** the dataset lacks order frequency, margin, tenure, business size, and other firmographic variables that could explain the underlying reasons for different spend patterns.
- 4. K-Means geometry:** K-Means is most naturally suited to compact, roughly spherical clusters under Euclidean distance. The current solution should therefore be treated as one useful segmentation view rather than a unique ground truth.
- 5. Log transformation:** the model operates in transformed standardized space, so distances in that space should not be translated directly into monetary differences. Raw means and medians are the correct basis for dollar-denominated interpretation.
- 6. Coarse segmentation:** K=2 is the selected stable solution, but its relatively low silhouette and strong post-hoc Channel alignment indicate that it captures a broad purchasing split and may miss finer segments.

8. Conclusion

The project identified two stable customer segments from six annual spending variables. K=2 was selected because it achieved the highest K-Means silhouette (0.290) in the tested K range and remained stable under both seed and bootstrap checks. The inertia curve does not show a uniquely sharp elbow, so it was treated as supporting evidence rather than an automated selection rule. K=3 and K=4 were not preferred once stability, separation, and interpretability were considered.

The two spend-based segments align strongly with the pre-existing Channel variable even though Channel was excluded from clustering. This is post-hoc contextual evidence of business relevance, not external validation. The result is therefore best interpreted as a broad segmentation of purchasing behavior rather than a definitive taxonomy of all customer types.

9. References

1. UCI Machine Learning Repository — Wholesale Customers: [Open source record](#)
2. Scikit-learn KMeans documentation: [K-Means documentation](#)

3. Scikit-learn silhouette_score documentation: [Silhouette score documentation](#)

Appendix: Reproducibility & Full Pipeline Reference

The analysis is implemented as phases 1 through 10. The final report manifest is recomputed from the source data and current intermediate outputs rather than from copied numbers. The pipeline uses K-Means with `random_state=42` and `n_init=20`. The primary clustering features are the six spend variables; Channel and Region are held out for post-hoc profiling. All major transformations, K-selection diagnostics, stability checks, cluster profiles, PCA diagnostics, and report-generation inputs are retained as intermediate artifacts.

Appendix: Reproducibility & Environment

Random seed:	42
K-Means:	n_init = 20
Preprocessing:	log1p + StandardScaler
Final K:	2
Stability:	Multiple random seeds + bootstrap ARI

Environment

Python:	3.13.5
pandas:	2.2.3
numpy:	2.3.5
scikit-learn:	1.8.0
scipy:	1.17.0
matplotlib:	3.10.8
seaborn:	0.13.2
JupyterLab:	4.5.3
Notebook:	7.5.3
pytest:	9.0.2

Reproduction note

The executed notebooks use the same canonical preprocessing and K-selection logic as the pipeline under src/. Re-running the pipeline with the pinned environment reproduces the documented analysis settings.